Mobile Squad Dashboard - Deployment Guide

Mobile Squad Dashboard - Complete Deployment Guide

Overview

This guide documents the complete process of deploying the Mobile Squad Repository Dashboard to GitHub Pages, from initial setup to live deployment.

Dashboard URL: https://blb-infios.github.io/mobile-squad-dashboard/
GitHub Repository: https://github.com/blb-infios/mobile-squad-dashboard
Auto-Update Schedule: Every 6 hours (4 times daily)

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Prerequisites

Before starting, ensure you have:

- ✅ Git installed and configured
- ✅ Node.js 18+ installed

- ✅ GitHub account with repository access
- ✅ Azure DevOps account (Highjumpvso organization access)
- ✅ Dashboard project code ready locally

Step 1: Project Setup

1.1 Project Structure

The Mobile Squad Dashboard project has the following structure:

```
mobile-squad-dashboard/
├── .github/
│   └── workflows/
│       ├── github-pages.yml          # Main deployment workflow
│       ├── azure-static-web-apps-deploy.yml
│       └── update-dashboard.yml
├── public/
│   ├── index.html                    # Dashboard UI
│   ├── styles.css                    # Styling
│   └── dashboard.js                  # Frontend logic
├── src/
│   └── aggregate.js                  # Data aggregation script
├── package.json
├── .env.example
├── .gitignore
├── README.md
└── QUICKSTART.md
```

1.2 Key Files

`.github/workflows/github-pages.yml` - Main workflow that: - Fetches data from Azure DevOps every 6 hours - Generates dashboard - Deploys to GitHub Pages

`src/aggregate.js` - Node.js script that: - Queries Azure DevOps REST API - Fetches repository data (builds, commits, PRs) - Generates `public/data.json`

`public/index.html` - Dashboard UI that: - Displays repository health status - Shows latest builds and commits - Auto-refreshes data display

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Step 2: Push Code to GitHub

2.1 Initialize Git Repository

```
cd /path/to/mobile-squad-dashboard
```

```
# Initialize git (if not already done)
git init

# Add all files
git add .

# Create initial commit
git commit -m "Initial commit: Mobile Squad Dashboard POC - Option 2"

# Rename branch to main
git branch -M main
```

2.2 Add GitHub Remote

```
# Add your GitHub repository as remote
git remote add origin https://github.com/blb-infios/mobile-squad-
        dashboard.git

# Verify remote
git remote -v
```

Expected output:

```
origin  https://github.com/blb-infios/mobile-squad-dashboard.git
(fetch)
origin  https://github.com/blb-infios/mobile-squad-dashboard.git
(push)
```

2.3 Push to GitHub

```
# Push code to GitHub
git push -u origin main
```

Expected output:

```
Enumerating objects: 23, done.
Counting objects: 100% (23/23), done.
Delta compression using up to 12 threads
Compressing objects: 100% (21/21), done.
Writing objects: 100% (23/23), 28.12 KiB | 9.37 MiB/s, done.
Total 23 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/blb-infios/mobile-squad-dashboard.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

2.4 Verify on GitHub

Go to: https://github.com/blb-infios/mobile-squad-dashboard

You should see: - ✅ All project files - ✅ README.md displayed - ✅ `.github/workflows/` directory with workflow files

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Step 3: Create Azure DevOps Personal Access Token

The dashboard needs a Personal Access Token (PAT) to read data from Azure DevOps.

3.1 Navigate to Token Creation Page

Go to: **https://highjumpvso.visualstudio.com/_usersSettings/tokens**

3.2 Create New Token

Click **"New Token"** button

3.3 Configure Token

Fill in the following details:

Name:

```
Mobile Squad Dashboard
```

Organization:

```
Highjumpvso
```

Expiration:

```
1 year (recommended)
```

Scopes: Select **"Custom defined"** and check these boxes:

```
☑ Code
  ☑ Read

☑ Build
  ☑ Read

☑ Work Items
  ☑ Read
```

DO NOT select: - Write permissions - Execute permissions - Other scopes

3.4 Create and Copy Token

1. Click **"Create"** button
2. **IMMEDIATELY COPY THE TOKEN** - you won't see it again!
3. Store it temporarily in a secure location (you'll use it in the next step)

Security Note: This token provides read-only access to your Azure DevOps repositories. Keep it secure and never commit it to git.

Step 4: Add Secret to GitHub

4.1 Navigate to Secrets Page

Go to: **https://github.com/blb-infios/mobile-squad-dashboard/settings/secrets/actions**

Or manually: 1. Go to your repository 2. Click **"Settings"** tab (top menu) 3. Click **"Secrets and variables"** (left sidebar) 4. Click **"Actions"**

4.2 Add New Secret

1. Click the green **"New repository secret"** button (top right)
2. Fill in:

   `Name: ADO_PAT`

   IMPORTANT: Must be exactly `ADO_PAT` (all caps, with underscore)
3. **Secret:** Paste your Azure DevOps token from Step 3
4. Click **"Add secret"** button

4.3 Verify Secret Added

You should see:

```
ADO_PAT    Updated now by you
```

Security Note: GitHub encrypts secrets and never displays them after creation. Only workflow runs can access them.

Step 5: Configure GitHub Pages

5.1 Understand the Process

Before configuring, the GitHub Actions workflow needs to run once to create the `gh-pages` branch. We'll trigger this first, then configure Pages.

5.2 Trigger First Workflow Run

Navigate to Actions: https://github.com/blb-infios/mobile-squad-dashboard/actions

Run the workflow: 1. Click **“Deploy to GitHub Pages”** in the left sidebar 2. Click the **“Run workflow”** dropdown button (top right) 3. Keep branch as **“main”** 4. Click **“Run workflow”** button

Wait for completion: - ⚪ Yellow dot = Running (wait 1-2 minutes) - ✅ Green checkmark = Success (proceed to next step) - ❌ Red X = Failed (see Troubleshooting)

5.3 Navigate to Pages Settings

After workflow succeeds, go to: **https://github.com/blb-infios/mobile-squad-dashboard/settings/pages**

Or manually: 1. Click **“Settings”** tab 2. Click **“Pages”** (left sidebar, under “Code and automation”)

5.4 Configure Source

On the Pages settings page:

Source dropdown: - Click dropdown - Select **“Deploy from a branch”**

Branch selection: - First dropdown: Select **`gh-pages`** - Second dropdown: Select **`/ (root)`**

Save: - Click **“Save”** button

5.5 Confirmation

You should see a blue box that says:

```
✓ Your GitHub Pages site is currently being built from the gh-pages branch.
```

Within 1-2 minutes, it will change to a green box:

```
✓ Your site is live at https://blb-infios.github.io/mobile-squad-dashboard/
```

Step 6: Trigger Initial Deployment

6.1 Wait for Pages to Build

After configuring GitHub Pages in Step 5, the site builds automatically.

Progress indicators: 1. **GitHub Pages settings page** shows: - Blue box: “Currently being built” - Changes to green: “Your site is live”

2. **Actions tab** shows:
    - New workflow run: “pages build and deployment”

- Status: Success ✅

6.2 Verify Build Status

Go to: https://github.com/blb-infios/mobile-squad-dashboard/actions

You should see two successful workflows: - ✅ "Deploy to GitHub Pages" (data fetch + deploy) - ✅ "pages build and deployment" (GitHub Pages build)

Step 7: Verify Dashboard is Live

7.1 Open Dashboard

Go to: **https://blb-infios.github.io/mobile-squad-dashboard/**

7.2 Expected Dashboard Content

Summary Cards (Top):

```
Total Repositories: 14
Healthy: 13
Warning: 0
Unhealthy: 1
Unknown: 0
```

Repository Sections:

1. KMP Libraries (6 repos): - Dex - K1V1 Authorization - Scanner (BarcodeScanning) - Printing - Beep - Vibrate

2. Capacitor Plugins (7 repos): - K1Authorization - Printing - Camera - Logger - Beep - MDM Reader - Vibrate

3. Security (1 repo): - MobileToolKit.Maintenance

7.3 Repository Card Details

Each repository card should display:

Header: - Repository name (clickable link to ADO) - Health status badge (color-coded)

Details: - **Last Build:** Status (succeeded/failed), build number, time ago - **Last Commit:** Commit ID (short hash), author, time ago - **Commit Message:** First line of commit message - **Active PRs:** Count (if any)

Timestamp: - "Last updated: [timestamp]" at top of dashboard

7.4 Verify Data Freshness

Check that: - ✅ Latest commits show recent work - ✅ Build statuses reflect current pipeline state - ✅ PR counts are accurate - ✅ Timestamps are current

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Troubleshooting

Issue 1: Workflow Failing - Missing ADO_PAT

Symptoms: - ❌ Red X on "Deploy to GitHub Pages" workflow - Error: "ADO_PAT environment variable is required"

Solution: 1. Verify secret exists: https://github.com/blb-infios/mobile-squad-dashboard/settings/secrets/actions 2. Should see `ADO_PAT` in the list 3. If missing, repeat Step 4 4. Re-run workflow

Issue 2: Workflow Failing - Invalid ADO_PAT

Symptoms: - ❌ Red X on workflow - Error: "401 Unauthorized" or "403 Forbidden"

Solution: 1. ADO PAT may have expired or incorrect scopes 2. Generate new token: https://highjumpvso.visualstudio.com/_usersSettings/tokens 3. Ensure scopes: Code (Read), Build (Read), Work Items (Read) 4. Update GitHub secret with new token 5. Re-run workflow

Issue 3: Dashboard Shows 404 Not Found

Symptoms: - URL returns "404 - Page Not Found" - GitHub Pages settings shows Pages is enabled

Solution: 1. Verify `gh-pages` branch exists: - Go to: https://github.com/blb-infios/mobile-squad-dashboard/branches - Should see `gh-pages` branch 2. If missing: - Workflow didn't complete successfully - Check Actions tab for errors - Fix errors and re-run 3. Wait 2-3 minutes after first Pages configuration 4. Clear browser cache and retry

Issue 4: Dashboard Shows Old Data

Symptoms: - Dashboard loads but data is stale - "Last updated" timestamp is old

Solution: 1. Check recent workflow runs: https://github.com/blb-infios/mobile-squad-dashboard/actions 2. Verify "Deploy to GitHub Pages" ran successfully 3. If last run failed, check logs and fix 4. Manually trigger workflow: - Actions → Deploy to GitHub Pages → Run workflow 5. Wait 1-2 minutes, then refresh dashboard

Issue 5: Some Repositories Show “Unknown” Status

Symptoms: - Repository cards show “Unknown” or “No Data” - Other repos work fine

Solution: - This is normal if repository has no builds/pipelines configured - ADO API returns no data for repos without pipelines - Repository still shows latest commit information - Consider setting up CI/CD pipeline for affected repos

Issue 6: GitHub Pages Branch Not Appearing

Symptoms: - Pages settings doesn’t show `gh-pages` branch option - Only shows `main` branch

Solution: 1. Workflow must complete successfully first 2. Check workflow status: https://github.com/blb-infios/mobile-squad-dashboard/actions 3. If successful, refresh Pages settings page 4. If still missing after successful run: - Wait 5 minutes - Clear browser cache - Try different browser

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Maintenance

Updating Dashboard Schedule

To change the auto-update frequency, edit `.github/workflows/github-pages.yml`:

```yaml
schedule:
  - cron: '0 */6 * * *'  # Current: Every 6 hours
```

Common schedules:

```yaml
# Every 3 hours
- cron: '0 */3 * * *'

# Every hour
- cron: '0 * * * *'

# Twice daily (9 AM and 5 PM UTC)
- cron: '0 9,17 * * *'

# Once daily at midnight UTC
- cron: '0 0 * * *'

# Weekdays only, every 4 hours
- cron: '0 */4 * * 1-5'
```

After changing, commit and push:

```bash
git add .github/workflows/github-pages.yml
git commit -m "Update dashboard refresh schedule"
git push origin main
```

Adding New Repositories

To track additional repositories, edit `src/aggregate.js`:

```
const REPOSITORIES = {
  kmp: [
    { name: 'Koerber.Mobile.Kotlin.Dex', displayName: 'Dex' },
    { name: 'Koerber.Mobile.Kotlin.NewRepo', displayName: 'New Repo'
        }, // Add here
  ],
  capacitor: [
    // ... existing repos
  ],
  security: [
    // ... existing repos
  ]
};
```

Commit and push changes:

```
git add src/aggregate.js
git commit -m "Add new repository to dashboard"
git push origin main
```

The workflow will automatically deploy the update.

Rotating Azure DevOps PAT

PATs expire after their configured lifetime. To rotate:

1. **Generate new token:** https://highjumpvso.visualstudio.com/_usersSettings/tokens
2. **Update GitHub secret:**
   - Go to: https://github.com/blb-infios/mobile-squad-dashboard/settings/secrets/actions
   - Click pencil icon next to `ADO_PAT`
   - Paste new token value
   - Save
3. **Test:** Manually trigger workflow to verify new token works

Recommended: Set calendar reminder 1 week before PAT expiration.

Monitoring Dashboard Health

Check workflow runs regularly: https://github.com/blb-infios/mobile-squad-dashboard/actions

Look for: - ✅ Green checkmarks (healthy) - ❌ Red X marks (needs attention) - ⚠️ Yellow warnings (review logs)

Set up notifications: 1. Go to: https://github.com/blb-infios/mobile-squad-dashboard 2. Click “Watch” → “Custom” → “Actions” 3. Get email on workflow failures

Backup and Recovery

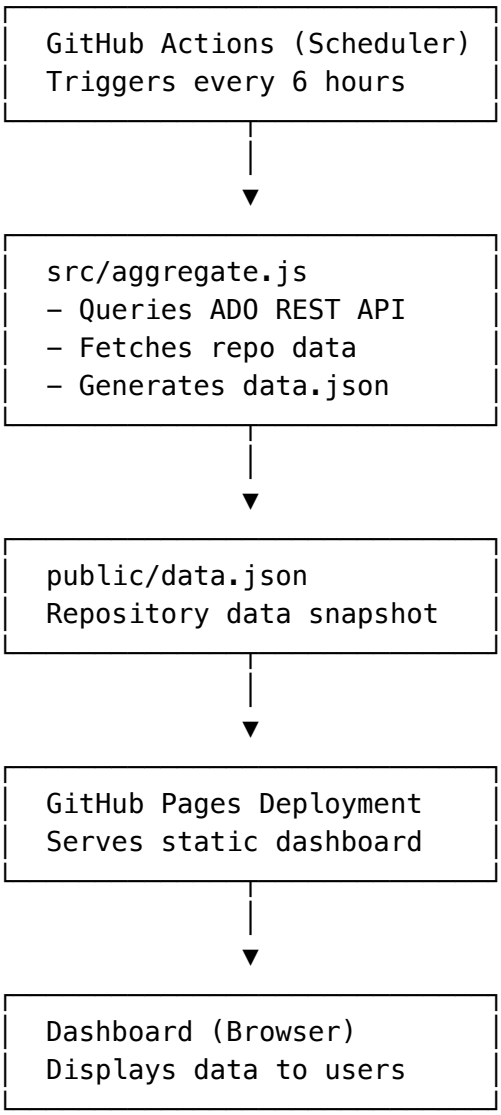
Backup strategy: - ✅ Code is version-controlled on GitHub - ✅ Secrets documented (regenerate if needed) - ✅ Configuration in git repository

Recovery steps if needed: 1. Clone repository 2. Regenerate ADO_PAT 3. Add secret to GitHub 4. Re-enable GitHub Pages 5. Trigger workflow

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Architecture Summary

Data Flow



Technologies Used

Component	Technology	Purpose
Backend	Node.js	Data aggregation script

Component	Technology	Purpose
API Client	Axios	Azure DevOps REST API calls
Frontend	HTML/CSS/JS	Dashboard UI
Hosting	GitHub Pages	Free static site hosting
CI/CD	GitHub Actions	Automated deployment
Scheduling	Cron (GitHub Actions)	Auto-refresh every 6 hours
Authentication	Azure DevOps PAT	Read-only API access

Cost Analysis

GitHub Pages

- **Cost:** $0/month (Free tier)
- **Included:**
  - 100 GB bandwidth/month
  - Unlimited builds
  - HTTPS/SSL certificates
  - Custom domains (if needed)

GitHub Actions

- **Cost:** $0/month (Free tier)
- **Included:**
  - 2,000 minutes/month (public repos)
  - Workflow runs: ~4/day × 2 min = 240 min/month
  - Well within free tier

Azure DevOps API

- **Cost:** $0 (read-only access)
- No API charges for read operations

Total Cost: $0/month ✅

Quick Reference

Important URLs

Resource	URL
Dashboard (Live)	https://blb-infios.github.io/mobile-squad-dashboard/
GitHub Repository	https://github.com/blb-infios/mobile-squad-dashboard

Resource	URL
GitHub Actions	https://github.com/blb-infios/mobile-squad-dashboard/actions
GitHub Secrets	https://github.com/blb-infios/mobile-squad-dashboard/settings/secrets/actions
GitHub Pages Settings	https://github.com/blb-infios/mobile-squad-dashboard/settings/pages
ADO Token Management	https://highjumpvso.visualstudio.com/_usersSettings/tokens
Azure DevOps Project	https://highjumpvso.visualstudio.com/WES

Key Files to Know

File	Purpose
`.github/workflows/github-pages.yml`	Main deployment workflow
`src/aggregate.js`	Data aggregation script
`public/index.html`	Dashboard UI
`public/data.json`	Generated data (updated by workflow)
`README.md`	Project documentation

Common Commands

```
# Test locally
cd mobile-squad-dashboard
npm install
npm run aggregate  # Requires ADO_PAT env var
npm run serve      # View at http://localhost:8080

# Update and deploy
git add .
git commit -m "Description of changes"
git push origin main  # Triggers auto-deployment

# Check deployment status
# Visit: https://github.com/blb-infios/mobile-squad-dashboard/actions
```

Success Criteria

✅ Deployment Successful When:

1. ✅ Code pushed to GitHub
2. ✅ GitHub Actions workflow runs successfully
3. ✅ GitHub Pages enabled and building

4. ✅ Dashboard accessible at: https://blb-infios.github.io/mobile-squad-dashboard/
5. ✅ All 14 repositories displayed
6. ✅ Health status showing (13 healthy, 1 unhealthy)
7. ✅ Latest commits and builds visible
8. ✅ “Last updated” timestamp current
9. ✅ Auto-update schedule working (check Actions tab)
10. ✅ No errors in workflow logs

Related Work Items

- **Feature #505124:** R&D - Mobile Squad Repository Monitoring Dashboard
- **PBI #506183:** POC Option 2 - Aggregation Script with Static Site ✅ **COMPLETE**

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